

Transaction Concept & ACID Properties — DBMS

Definition: A **Transaction** is a logical unit of database processing that includes one or more database operations (READ, WRITE). To preserve data integrity, transactions must satisfy all four **ACID Properties**.

1. Detailed Technical Explanation

The ACID Properties:

A - ATOMICITY : All operations complete successfully, or ALL are undone ("All-or-Nothing").
C - CONSISTENCY : Transaction transforms database from one valid state to another valid state.
I - ISOLATION : Concurrent transactions execute independently without interfering with each other.
D - DURABILITY : Once committed, updates persist permanently in non-volatile storage even after system failure.

1. Atomicity

- Enforced by the **Recovery Manager** using Write-Ahead Logging (WAL) and Rollback mechanisms.
- Operations: **COMMIT** (permanent save) or **ROLLBACK** (undo changes).

2. Consistency

- Enforced by database schema constraints (Primary Key, Foreign Key, **CHECK** constraints) and application logic.

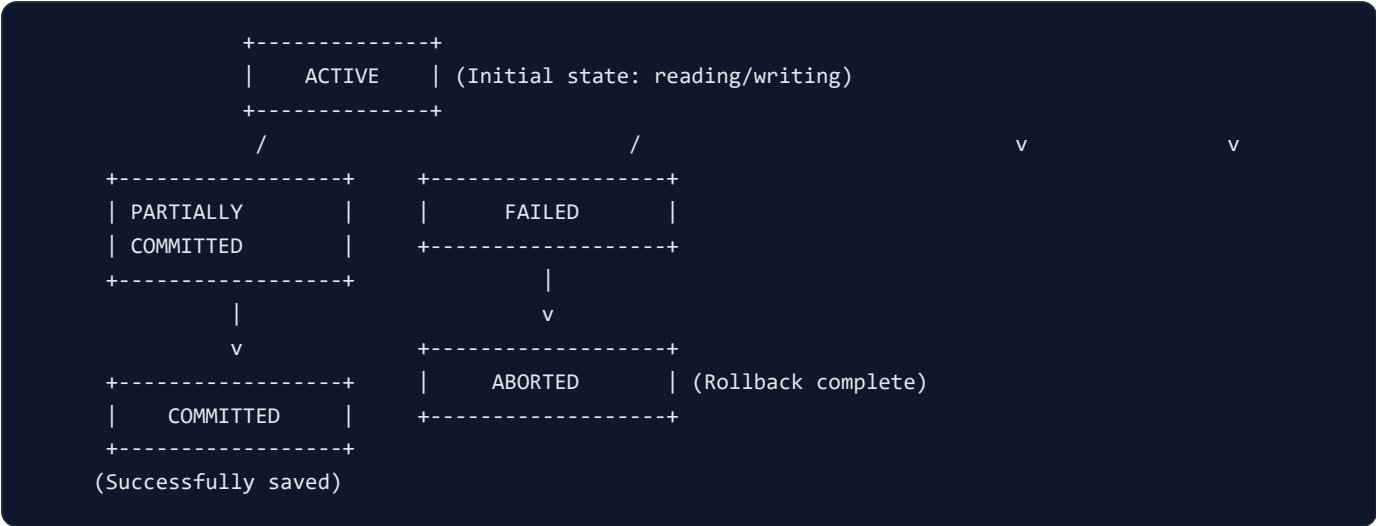
3. Isolation

- Enforced by the **Concurrency Control Manager** using locking or timestamping.
- Prevents anomalies: Dirty Reads, Non-Repeatable Reads, Phantom Reads.

4. Durability

- Enforced by **Log Recovery & Checkpointing** writing updates to non-volatile disk/SSD storage.

2. Transaction State Diagram



3. Core Concepts & Memory Keywords

- **ACID:** Atomicity, Consistency, Isolation, Durability.
- **Commit Point:** The exact moment transaction updates become permanent.
- **Rollback / Abort:** Undoing all operations of an incomplete transaction.

4. Must-Write Points for Exams

- Atomicity guarantees that partial execution of a transaction is impossible.
- Isolation ensures that concurrent execution yields the same state as serial execution.
- The transaction state transitions from Active -> Partially Committed -> Committed, or Active -> Failed -> Aborted.

5. Quick Recall Flow

Active -> Read/Write -> Partially Committed -> Commit Log Written -> Committed (Or Aborted on Failure)